Cayley's Collected Mathematical Papers, Vol. II
Pages 285–334 (PDF 301–350)
Paper 143 (concluded) and Paper 144 (opening)

143. Tables of the Covariants M to W of the Binary Quintic (concluded).

[From the Philosophical Transactions of the Royal Society of London, vol. cxlvi. for the year 1857, pp. 703–755.]

The present pages contain the remaining tables of the covariants of the binary quintic, namely tables N to W, followed by the verifications of the numerical coefficients. The covariants are linear forms in the powers of the leading coefficient a, with coefficients which are themselves polynomial in b, c, d, e, f multiplied by powers of x and y. Each table is laid out as a multi-column array of monomial-coefficient pairs. The original printed layout, with its column alignment, is preserved here in verbatim blocks; the OCR-derived text retains the typographical features of the 1889 edition (subscripts and dashes appearing as encoded by the OCR engine), and is to be read as a faithful skeleton rather than as a corrected re-typesetting.

Reading convention. A line such as "$a^2b^4cdf - 1$" means: the term a^2b^4cdf, in the relevant covariant, has numerical coefficient -1. The covariants are arranged column by column; each column corresponds to a power of a, the leading coefficient, and within each column the monomials are listed in the natural lex order. At the foot of each column is the algebraic sum of the (positive minus the negative) numerical coefficients in that column; at the foot of each table is the total numerical sum, which agrees with the sum at the foot of the column for the lowest power of a and which provides the principal numerical check.

N. No. 84.

[The full coefficient table occupies book p. 285; it lists the terms of the covariant N of the binary quintic, arranged in five columns headed by the powers a^6, a^5, a^4, a^3, a^2 of the leading coefficient. The column-sums are -19, $+81$, -62, ± 0, $+62$; total ± 163. Owing to its size and the OCR damage to subscripts in this particular table, the table is reproduced below in literal OCR form; the reader is referred to the scan for definitive values.]

```
a^6 b^7df  -1     a^5 b^4ef        a^4 b^4cf         a^3 b^4f          a^2 b^5f^2 +1
   cdf  +1        a^5 b^3df -4        b^3ef              b^3cef +4         def^2 -3
a^5 b^4df^2 +3       def +4           b^3cdf            def^2 -4         ey +2
                                      a^4 b^3df^2 -6       b^3cef +8     b^4cdf^2 -2
b^4cdef -2           a^4 b^3df^2 +4                      def +12          d^2ef +8
                                      b^3cef +12                          d^2e -2
b^3def +12           def +12          cdef             cef^2 -32          c^2e^2 +6
cd^2f +2                                                                  cd^2e -20
cd^2 +12             cd^2f +16        a^4 b^3df^2 +14    cef +6            cde^2 +12
                     e^4 -24                                                 cd^2 -2
   d^2e -6           b^2c^2ef -88   c^2f^2 +13    b^3c^2ef +56
                                                                           a^3 b^4f^2 +5
a^4 b^3f^2 -2        cd^2 +30         b^2c^2df^2 +48
                                                                              dey -13
   b^2cef +2         a^3 b^3df^2 -4     c^2def -88                            ef +12
   d^2f -6                                cd^2 +56                         b^4cef^2 -12
                                                                              c^2ef -13
   bce^2 +4     a^3 b^3cdf -52        a^3 b^3df^2 -6  cd^2 +12                d^2ef -4
   cd^2e -10         d^4 +24          a^3 b^3def -56      cdef -40          ef^2 +24
                                                                               cdef +28
   a^3 b^3f^2 +6    b^4ef -9          def +36          e^4 -60                cf^3
                                                                              ce^2 -15
   bcdf +12          cd^2e +20          b^3cdf +36    b^3cef +24              d^2ef -4
   b^3ef -15         b^4cdef -10       c^2def -156    cdef -56                c^2ef +52
```

```
                                                                            d^2e^2 -10
d^3e +10         d^2 -100        e^2def -50       c^2dey -20              de^2 -20
a^4 b^3ef +6                                                               c^2de^2 +30
bcdf +12         a^3 b^3cef -12  b^3cef +18       c^2de^2 +100
c^2ef -15                          cd^2 -156                               d^4e +15
```

...

[Reproduced for the entire breadth of the table from the OCR layer; alignment artefacts reflect the broken sub/superscript glyphs in the scan. Column totals at foot: -19; $+81$; -62; ± 0; $+62$. *Grand total:* ± 163.*]*

O. No. 90.

[Book p. 286. The covariant O has degree higher than N. The terms with monomials of the form $a^p b^q c^r d^s e^t f^u$ *are arranged in four columns headed by descending powers* a^6, a^5, a^4, a^3 *of a. Column sums:* 4, 59, 497, 1003*; column counter-sums:* 1, 49, 669, 964*; differences in each column,* ± 1563.*]*

```
a^6 b^4f^2 +1      a^5 b^4df^2 -1      a^4 b^4f^2        a^3 b^4ef +15
      def -4          ey^2 +1             def            -b^4cd^2 -26
    ey +3          a^5 b^4cf^2 +4                            cey -84
    a^5 b^4f^2 -1                         a^4 b^4ef +7      d^2ef +98
    b^4cef -3          dey +3                b^4df -30
                       cey -7                                -de^4 -45
       dey +16     b^3cef -16               cey +1           b^4ef +12
                                            d^4ef -74
       dey +4         cd^2f +6                               c^2def +94
                                            cd^2 +84
       e^4 -15        cdey +30              b^4cdef +18      c^3 +150
    b^3cdf^2 -6       ce^4 -8                                cdy -140
                                              cdef +160      cd^4 -50
       c^2ef +4       d^3 -18               -cd^4e^2 32
       cd^2ef -22     +ddey -6
                                            d^4f +81         d^4e +15
                                            ce^4 +6          b^4c^2 -51
       cd^2 +26       a^3 b^4f^2 -3
                                            b^4c^2 -9        c^3df +100
       d^4f +9        b^4cef^2 -4
       d^2ey -12      dey^2 -4              c^2def +20       -c^4ey 320
                       d^2ey -1
    a^3 b^4ef +7                            c^2e^2 -112      c^4e +310
       b^4d^2 -30        ey +18             c^3dey -18
                                                             cd^4 -90
       cey +1          b^4cdf^2 +22         c^3e^2 +284      b^4cdf -18
       d^2ef -74       c^2d^2f +74          cd^4 -216
                                                              c^2 +120
       d^4e +84        cdey^2 -160          d^4 +54          c^3d^2e -130
       b^4cdf^2 +18    -cd^4e^2 32          a^3 b^4ef +15    c^3d^4 +40
           cdef +160     d^4f +81           -b^4cdf^2 26
       -cdye^2 150       cdey +6            cey -84           :  4              1
                           b^4cdf -9                                59         49
                                              d^4ef +98                       669
```

[OCR layout preserved; sub/superscripts as in the scanned printing.]

P. No. 91.

[Book p. 287. The covariant P is arranged in six columns by powers a^6, a^5, a^4, a^3, a^2, a. *Column sums:* -3, $+67$, ± 6, ± 6, ± 27, -1*; sub-sums per row given at the right; grand totals* ± 388 *and* ± 594 *for the two halves* x *and* y *of* $\square\, x, y \square^k$.*]*

```
a^6 b^4f^2     a^5 b^4f^2     a^4 b^4f^2 -1     a^3 b^4df^2 +1     a^2 b^4ef^2     a b^4f^2
                                       dey^2 -1
a^5 b^4ef^2    -b^4cdef 2     def^2 +6                             a^2 b^4df^2 +2    a^5 b^4ef^2
                                                 a^4 b^4cdef -6
   b^4cdf +1   cd^2f +5        ey -5                                ey^2 -2           b^4cd^2 -1
                cdy -1                                                                  cey +1
     cey -2                    by^2 +1           def^2 +11          b^4c^2f -5
                                                                                        cf^2 +3
cd^2ef +2      e^4 -2          -b^4cdy^2 11     ey -5               cy^2 +17          dey -5
                                d^4f -4                             cey -7
cd^4 -1        b^4cy +2        d^4ef -4          b^4c^2f^2 +4       cy^2 -4           ey +2
                                                   cdy^2 -2
a^4 b^4dy^2 -1 -b^4cdf^2 17                                                          a^3 b^4dy^2 +2
                                                                    d^4ey -6
   ey +2        cey +13         e^4 +17         c^4ey +4                                 ey^2 -2
                d^4ef -32                                           dey +5
b^4cy^2 -3                        b^4c^2dy^2 +2  ce^4 -4                              b^4cy^2 -2
                                                 d^4ef -10          a^3 b^4cf^2 +1
   cdef -6      dey +32             c^4ey +26                      -dey^2 13          c^4y^2 +6
ce^2 +13       b^4cy^2 +4                        d^4e^2 +8           ey +12             cey -2
                                    cd^4ef -2                                         cdy -16
c^2y -8         c^2dy +36       -cd^4y 40        a^3 b^4y^2 +5      b^4cey^2 +32
                cy -24                                                                d^4ey +24
                                    d^4f -9      b^4cey^2 +4         cdy^2 -36        dey -10
+d^4e^2 2      cd^4f -10
                                    +-dey^2 24   dy^2 -26            cdey -42
b^4c^4f +16   cd^4e -16                           dey -35              cey +24         b^4cdy^2 +8
                                a^3 b^4y^2 +5                                           c^4y^2 +2
   cd^4ef -2                                     e^4 +42             ce^4 +24
                d^4e +12        b^4cdy^2 -4       b^4c^4df^2 +2      d^4ey +56          c^4dy -52
   c^3dey -38                                                        d^4e^2 -34
                a^4 b^4cdef +7      cey +35      c^4ey +26                               c^4y +28
        c^4e +34                    d^4ef -26                                                  ...
```

[Continues for several columns; total numerical sums in the source printing: ±388, ±594 for the x- and y-halves of the binary form.]

Q. No. 25, Q′. No. 26.

[Book p. 288. The invariants Q and Q′. The table is arranged with two adjacent columns labelled Q and Q′, listing the same monomials in a, b, c, d, e, f with the corresponding signed integer coefficients. The numerical sums for the powers a^5, a^4, a^3, a^2, a of a are listed at the right edge of the table.]

```
                                  Q            Q'

a^6 b^4f^4                       +1            -3375
   b^4cdef                                       -20
   b^3cdf^2       +1             -120
   c^3f           -1
   c^3e           -3              +160
   d^2ef^2        +5              +360
   d^4f           -1              -640
   e^4            +1              +256

a^5 b^4ef^3                      +160
   b^3df^3                       -10                     (sub-totals
   cdef^2          +11                                    for Q':
   c^4f            +12           +360                     1          =1
   d^4f^2          -30                                   776-780    =-4
   d^2ey           +15          -1640                    21266-21250 =+6
   de^4            +12           +320                    68656-68660 =-4
```

	-21	-1440	37816-37815 =+1
b^4c^2f^2	-34	+4080	128505-128505 = 0)
c^4ef^2	+22		...
	+78	-1920	
c^3def	-48	-1440	
c^4y	-27	-1440	
cd^2ef	+18	+2640	
cd^4y		+4480	
d^4f	+5	+2640	
d^3e^2	-5	-2560	
		-10080	
a^4 b^4cf^2	-30		
def	-34	+5760	
ey		+3456	
	+133	-2160	
b^4c^4ef	-54	-640	
cd^4y	-18		
		+320	
cde^4f	+3	-180	
	+78		
ce^4	-18	+4080	
	-220	+4480	
d^4ef	+106	-14920	
	+93	+7200	
d^4e^2	-30	+960	
	-9	-600	

[The full table runs across two columns (Q and Q') for several screens; at the end of the table the column sums are ± 6 for Q and ± 128505 for Q'. The sub-totals $169, 625, 424$ accumulate to ± 1124 for the partial-sums check.]

R. No. 92.

[Book p. 289. The covariant R. Six columns headed by powers a^6, a^5, a^4, a^3, a^2, a of a; the column sums are ± 8, 93, 300, 780, totalling ± 1181.]

a^6 b^4ef^2	a^5 b^4cdy -15	a^4 b^4f^2	a^3 b^4e^4	a^2 b^4f	a b^4cdef +2
	cd^2 -38	a^3 b^4ef^2			
a^5 b^4df^2	de^4 +45	b^3cdf^2 +18	b^4cdf^2 +18	b^4cef	d^2e^2 +15
		c^4cf -66			
cd^2	b^4cf +3	c^4f	cd^4ef +20	d^4f +"i"	b^3cdf^2 -32
			cd^4		
b^3c^4f -1		d^4ef +2		de^4f -2	c^4cf -39
	cd^4ef +102		d^4f +58	e^4f +1	
cd^4f +6					c^4def -24
	c^4 -15	d^4f -4		a^2 b^4ef^2	
c^4f -4			b^4c^4f -50		c^4de^2 +17
	cd^4f +76	d^4 +2	b^3c^4f -6	b^3cdf^2 -6	
c^4f^2 -3					cd^4f +25
	cd^4 -175	a^3 b^3df	c^4def +72	ce^4f +6	cd^4e^2 -1
d^4e^2f^2	d^4e +35	e^4	c^4 +50	d^4ef^2 +3	
				d^4ey	d^4e +15
de^4 +1	b^4c^4ef -42	b^3cf -2	d^4f -3		
				e^4 -3	b^4f^2 +9
a^5 b^4cf^2 +2	c^4d^4f -182	cdef	c^4d^4f -156		
def^2 -6				b^4cf^2 +3	c^4ey +106
	c^4d^4 +120	cey +4	cd^4e^2	c^4def -3	c^4 -35
e^4f +4	c^4d^4 +150	d^4ey -14			c^4d^4f -
b^3c^4y -3		d^4ey +30	cd^4e +90	cd^4y -6	
cdf^2 +3	cd^4 -70	de^4 -18	d^4 -30	cd^4f	cd^4e^2 -
	b^4cdf +126		b^4cef -24		cd^4e +175

```
cd^4f -18                                  b^4cdy^2 +14  c^4d^4y^2 94              cd^4ey^2 +3    cd^4 -45
                  -c^4e^4 15                                                       cde^4 +6     b^3c^4ef -3
ce^4 +17           c^4d^4e -175    c^4d^4y                                           d^4ef
                   c^4d^4 +75     c^4d^4ef -66       c^4d^4e -90                 -d^4e^2 3      c^4d^4f +21
   d^4ef +22       b^4c^4f -27    c^4 +26          c^4d^4e                   a^3 b^4df^2 +4
   d^4 -21                                                                                      + c^4d^4e^2
                                       cd^4ef +56     c^4d^4 +10                 e^4f -4
b^4cdy            + c^4de +45        cd^4 -18        b^4cdef -18                 b^4cf -1        c^4d^4e -7
                  -c^4de 20        c^4f -18          c^4 +30                       cdey^2 +18
e^4f +13                                                                           cey^2 -16      c^4d^4 +20
                                                                                                  ...
```

S. No. 93 bis; S′. No. 93. $(\diamond x, y)^4$.

[Book p. 290. The covariant pair S, S' is given as a binary quartic in x, y with five coefficients, one for each of $x^4, x^3y, x^2y^2, xy^3, y^4$. The contents of each coefficient (a homogeneous polynomial in a, b, c, d, e, f of the appropriate degree-weight) are tabulated side by side under headings S and S'. Per the printed table, the column sums of the numerical coefficients vanish for each separate power of a, in accordance with the verification rule derived from the Tenth Memoir on Quantics.]

```
Coef. x^4         S         S’      Coef. x^3 y        S         S’

a^4 b^4f^2                -9      a^3 b^4ef       -66    +528      a^4 b^3cf^2           +9    a^3 b^3c^4e^4 +6
                                                         -45                                         cd^4ef
```

[Table extends across all four coefficients $x^4, x^3y, x^2y^2, xy^3, y^4$ of the covariant S and the like for S'. Verifications consist in checking that $\sum$(positive) $= \sum$(negative) for every power of a.]

T. No. 94.

[Book pp. 291–293. The covariant T. As stated in the prose passage that follows, for T the sums are taken for the different combinations of a and b; the tabulation is two-column, the left column giving the x-coefficient, the right column the y-coefficient, of the covariant. Sample column-totals: for a^3b^5, the x-coefficient column has positive–negative $307 - 349$, $1073 - 1003$, $2040 - 2110$, $1930 - 1880$, $1207 - 1221$, $231 - 239$; for the y-coefficient column, the totals are 112, 281, 1650, 6511, etc., summing to ± 20196 each side. The full table occupies three book pages and is reproduced below in OCR-skeleton form.]

```
                  x coefficient.                              y coefficient.

     a^4 b^7                26          a^4 b^7           db  12

     a^3 b^6             - 14            26  a^3 b^6               2          12
          b^5                            436        b^5        112          395
          b^4                  141      3738                     281
                                        9116         b^4
     a^3 b^4                 281
                                        6880    a^2 b^7
       b^3                           -
                                                    b^6
          b^2               -1                      b^6          42
          b                ...                      b^5          546
                             106                               696
       b^7                                       a^3 b^5         366
                             186
     a^2 b^6                                        b^4
                             1173                   b^3
          b^5                                       b^2
```

```
                              2272                b
      b^4                                         b^0
      b^3                   - 16                                       5        1650
                                              a^2 b^4                179        6511
         b^2                   359                b^3              821
         b^1                  1411                                2097
                             3103                 b^2             2147
    a^1 b^3                   3030                 b^1             1262
                             1197
       b^2                                             b^0
                           -  2
         b^1                                       b^0
         b^0       92 --      78                   b^0                 28
         b                 -                                       342
                  307 -- 349                                      1790
            b^0                                                   3496
                  1073 -- 1003                                    3445
         b^0                                                       2064
                  2040 _ 2110                                      463
            b^0
                  1930 -- 1880

                  1207 -- 1221

                  231        239

                                                                          11628

                                       * 20196                      * 20196
```

Here for the x-coefficient: in the terms in a^4 the successive sums are $-2,\ +14,\ -42,\ +70,\ -70,\ +42,\ -14,\ +2$, which are the coefficients of $-2(\theta-1)^7$.

U. No. 95 (continued and concluded).

[Book pp. 294, 297, 298, 299, 300, 301–303. The covariant U extends across many book pages, in two parts (x-coefficient and y-coefficient). The column sums (taken over separate powers of a) for the x-coefficient on book p. 295 are: ±36, 198, 242, 2, 208, 286, 866, 1246, 64, 328, 1258, 2586, 2186, 856, 4, 70, 448, 1488, 2140, 167 *totalling* ±17264*. For the x- and y-coefficients of V (book p. 298), the column totals are* ±4, 76, $2956-3040$, $11946-11806$, $23924-24064$, $25110-25026$, $25524-25552$, $8822-8812$, *totalling* ±98358 *and* ±154122*. For W (book p. 299), the term in* a^4 *has successive sums* $-28,\ +84,\ +140,\ -140,\ +84,\ -28,\ +4$, *the coefficients of* $-4(\theta-1)^6$*. The full term-by-term tabulations are reproduced in the scan; for length reasons only the table-titles and verification sums are reprinted here.]*

V. No. 95 (continued). *y* coefficient.

[Book pp. 297–298. Format identical to the preceding. For details consult the scan.]

W. No. 29 A (continued and concluded).

[Book pp. 299–303. The covariant W. The largest table in the entire series of papers 141–143, occupying five book pages. Column sums for the a^4 *terms produce successive sums* $1,-10,+45,-120,+210,-252,+210,-120,+45,-10,+1$ *(i.e. the binomial coefficients of* $-(\theta-1)^{10}$*); for* a^3 *they are* $-5,+54,-265,+780,-1530,+2100,-2058,+1440,-705,+230,-45,+4$*, i.e. the binomial coefficients of* $-5(\theta-1)^{11}-(\theta-1)^{10}$*. The grand total of the table on book p. 303 is* ±2207351*, with the* $\pm$ *partial sums* ±2972759 *on the second part and* ±3111879, ±3003111, ±9087749 *on subsequent parts.]*

Verifications of the Numerical Coefficients (Book pp. 304–309).

For the lower covariants the numerical verifications are given for the entire coefficient, but for the higher ones where the number of terms in a coefficient is considerable they are given separately for the different powers of a; and it is also interesting to consider them for the separate combinations of a and b. I recall that the positive and negative numerical coefficients are summed separately, so that "$\pm$ (a number)" means that the sum of the positive numerical coefficients is equal to the sum of the negative numerical coefficients and thus that the whole sum is $= 0$.

It is to be observed that for the lower covariants the sums of the numerical coefficients do not vanish for the separate powers of a: thus in the invariant G, 141, the sums of the numerical coefficients for the terms in $a^4,\ a^3,\ a^2$ are $= 1,\ -2,\ 1$ respectively.

As regards the invariants Q and Q': for the first of these, Q, the sums of the numerical coefficients for the terms in $a^5,\ a^4,\ a^3,\ a^2,\ a$ are each of them $= 0$, but this is not the case as regards Q'; in fact $Q' = G^2+$ a multiple of Q; hence the sums for Q' are the same as those for G^2, viz. they are $= 1,\ -4,\ +6,\ -4,\ +1$ respectively. Like results present themselves in other cases, and they might probably be accounted for in a similar manner; we have a series of sums not each $= 0$, but which are equal to a set of binomial coefficients taken with the signs $+$ and $-$ alternately, and thus the sum of these sums is $= 0$.

For R, S and S', I have given the sums for the different powers of a; and in regard to S I give here the following paragraphs from the Tenth Memoir on Quantics:

> "I remark that I calculated the first two coefficients S_0, S_1 and deduced the other two, S_2 from S_1, and S_3 from S_0, by reversing the order of the letters (or which is the same thing, interchanging a and f, b and e, c and d) and reversing also the signs of the numerical coefficients. This process for S_2, S_3 is to a very great extent a verification of the values of S_0, S_1. For, as presently mentioned, the terms of S_0 form subdivisions such that in each subdivision the sum of the numerical coefficients is $= 0$: in passing by the reversal process to the value of S_3, the terms are distributed into an entirely new set of subdivisions, and then in each of these subdivisions the sum of the numerical coefficients is found to be $= 0$; and the like as regards S_1 and S_2."
>
> "If in the expressions for S_0, S_1, S_2, S_3 we first write $d = e = f = 1$, thus in effect combining the numerical coefficients for the terms which contain the same powers in a, b, c, we find"

$$\begin{aligned}
S_0 = {}& a^4(-2c^2 + 6c - 6c + 2) \\
& + a^3\,\{b^2(6c^2 - 12c + 6) + b(-15c^3 + 33c^2 - 21c + 3) \\
& \qquad + b^0(42c^4 - 147c^3 + 195c^2 - 117c + 27)\} \\
& + a^2\,[b^4 \cdot 0 + b^3(30c^2 - 36c + 6) + b^2(-117c^3 + 249c^2 - 183c + 51) \\
& \qquad + b(9c^4 + 148c^4 - 378c^3 + 330c^2 - 99c) + b^0(-63c^5 + 165c^4 - 147c^3 + 45c^2)]
\end{aligned}$$

and continuing on the next book page,

$$\begin{aligned}
& + a\,\{b^5 \cdot 2 + b^4(-15c + 3) + b^3(75c^2 - 69c + 24) + b^2(-9c^3 + 167c^2 - 225c + 87c - 2) \\
& \quad + b(72c^4 + 48c^3 - 186c^2 + 96c) + b^0(-126c^5 + 201c^3 - 87c^4) \\
& \quad + b^0(27c^6 - 40c^5 + 20c^4)\}
\end{aligned}$$

which for $c = 1$ becomes $= 2b^6 - 12b^5 + 30b^4 - 40b^3 + 30b^2 - 12b + 2$, that is $-2(b-1)^6$, and for $b = 1$ becomes $= 0$.

$$
\begin{aligned}
S_1 = \;& a^4(0c^2+0c+0)\\
&+a^3\{b^0(0c+0)+b(3c^3-9c^2+9c-3)+b^2(24c^4-99c^3+153c^2-105c+27)\}\\
&+a^2\{b^4\cdot 0+b^3(-6c^2+12c-6)+b^2(-24c^3+90c^2-108c+42)\\
&\qquad+b(33c^4-90c^3+54c^2+30c-27)+b^0(-27c^5+78c^4+66c^3+6c^2-9c)\}\\
&+a^0\{b^5(3c-3)+b^4(-15c+15)+b^3(60c^2-12c^3+36c+30)\\
&\qquad+b^2(9c^4-42c^3+84c^2-108c+57c)+b(0c^5-54c^4+96c^3-51c^2)\\
&\qquad+b^0(9c^6-9c^4)\}
\end{aligned}
$$

which for $c=1$ becomes $=0$.

$$
\begin{aligned}
S_2 = \;& a^4(0c+0)\\
&+a^3\{b^4\cdot 0+b^3(0c^2+0c+0)+b^2(18c^4-72c^3+108c^2-72c+18)\}\\
&+a\{f(0c+0)+b^3(-33c^3+99c^2-99c+33)+b(57c^4-162c^3+144c^2-30c-9)\\
&\qquad+b^0(-60c^5+207c^4-261c^3+141c^2-27c)\}\\
&+a^0\{b^5\cdot 0+b^4(15c^2-30c+15)+b^3(-54c^3+102c^2-42c-6)\\
&\qquad+b^2(123c^4-297c^3+243c^2-87c+18)+b(-27c^5+102c^4-96c^3+21c^2)\\
&\qquad+b^0(27c^6-60c^5+51c^4-12c^3)\}
\end{aligned}
$$

which for $c=1$ becomes $=0$.

$$
\begin{aligned}
S_3 = \;& a^4\cdot 0\\
&+a^3\{b(0c+0)+b^2(0c^3+0c^2+0c+0)\}\\
&+a^2\{b^3\cdot 0+b^2(0c^3+0c+0)+b(-9c^4+36c^3-54c^2+36c-9)\\
&\qquad+b^0(36c^5-171c^4+324c^3-306c^2+144c-27)\}\\
&+a^0\{b^4(0c+0)+b^3(7c^3-21c^2+21c-7)+b^2(-39c^4+135c^3-171c^2+93c-18)\\
&\qquad+b(66c^5-243c^4+333c^3-201c^2+45c)\\
&\qquad+b^0(-27c^6+101c^5-141c^4+87c^3-20c^2)\}
\end{aligned}
$$

which for $c=1$ becomes $=0$.

It follows that for $c=d=e=f=1$, the value of the covariant S is $=2(b-1)^6y^4$, which might be easily verified.

For T, U, and W, I look at the sums for the different combinations of a and b. The tables of these sums (book pp. 306–309) are reproduced verbatim above in the headings under T, U, V and W; only the principal verification identities are quoted here:

For T (book p. 306): the x-coefficient successive sums for a^4 are $-2, +14, -42, +70, -70, +42, -14, +2$, the binomial coefficients of $-2(\theta-1)^7$; the total ± 20196 on each side.

For U (book p. 307): successive sums vanish appropriately. Total ± 17264.

For V (book p. 308): the x-coefficient sums for a^4 are $-4, +28, -84, +140, -140, +84, -28, +4$, the coefficients of $-4(\theta-1)^7$; for a^3, the sums are $18, -140, +476, -924, +1120, -868, +420, -116, +14$, the coefficients of $18(\theta-1)^8+4(\theta-1)^7$. In the y-coefficient the successive sums are $-2, +16, -56, +112, -140, +112, -56, +16, -2$, the coefficients of $-2(\theta-1)^8$. Totals ± 98358, ± 154122 and ± 186734, ± 126760.

For W (book p. 309): For the terms in a^4 the successive sums are

$$1,\ -10,\ +45,\ -120,\ +210,\ -252,\ +210,\ -120,\ +45,\ -10,\ +1,$$

which are the coefficients of $-(\theta - 1)^{10}$; and for the terms in a^3 the successive sums are

$$-5,\ +54,\ -265,\ +780,\ -1530,\ +2100,\ -2058,\ +1440,\ -705,\ +230,\ -45,\ +4,$$

which are the coefficients of $-5(\theta-1)^{11}-(\theta-1)^{10}$. Totals ± 2207351, ± 2972759, ± 3003111, ± 3111879, ± 9087749.

[End of paper 143.]

144. A Third Memoir upon Quantics.

A Third Memoir upon Quantics.

[From the Philosophical Transactions of the Royal Society of London, vol. cxlvi. for the year 1856, pp. 627–647. Received March 13,—Read April 10, 1856.]

My object in the present memoir is chiefly to collect together and put upon record various results useful in the theories of the particular quantics to which they relate. The tables at the commencement relate to binary quantics, and are a direct sequel to the tables in my Second Memoir upon Quantics, vol. cxlvi. (1856), [141]. The definitions and explanations in the next part of the present memoir are given here for the sake of convenience, the further development of the subjects to which they relate being reserved for another occasion. The remainder of the memoir consists of tables and explanations relating to ternary quadrics and cubics.

Covariant and other Tables, Nos. 27 to 50 (Nos. 1 to 50 binary quantics).[1]

Nos. 27 to 29 are a continuation of the tables relating to the quintic

$$(a, b, c, d, e, f \mid x, y)^5.$$

No. 27 gives the values of the different determinants of the matrix

$$\begin{pmatrix} a, & 4b, & 6c, & 4d, & e & \cdot \\ \cdot & a, & 4b, & 6c, & 4d, & e \\ f, & 4c, & 6d, & 4e, & f & \cdot \\ \cdot & f, & 4c, & 6d, & 4e, & f \end{pmatrix}$$

determinants which are represented by 1234, 1235, &c., where the numbers refer to the different columns of the matrix. No. 28 gives the values of certain linear functions of these determinants, viz.

$$\begin{aligned} L &= +12.56 - 2345 - a \cdot 1346, \\ L' &= 3 \cdot 1256 - 1346, \\ 8M &= -1345 + 2 \cdot 1246, \\ 8M' &= -2346 + 2 \cdot 1356, \\ 8N &= -1245 + 3 \cdot 1236, \\ 8N' &= -2356 + 3 \cdot 1456, \\ 8P &= 2L' - 3L = 6 \cdot 1846 - 3 \cdot 2345, \\ 16P' &= -5L' - L = -16 \cdot 1256 - 3 \cdot 1346 - 2356. \end{aligned}$$

At the end of the two tables there are given certain relations which exist between the terms of Tables 14, 16, 25, 26, 27 and 28.

No. 27.

[Eight columns, one for each of the determinants 1234, 1235, 1236, 1245, 1246, 1345, 1256, 2345 (continued on the next page with 1346, 2346, 1356, 2356, 1456, 2456, 3456). Each column lists the monomial terms of a, b, c, d, e, f appearing in the expansion of the corresponding minor, together with the signed integer coefficients.]

1234.	1235.	1236.	1345.	1246.	1345.	1256.	2345.
a^3bf ...	a^3a^2 - 4	a^2df + 6	a^2df - 6	a^2cf + 4	a^2cy	a^2y^2 + 1	a^2cf
						abef -- 2	
a^2ce - 16	a^2de + 24	a^3 ...	a^2cf + 16	abdf - 4	ab^2y - 24	acdf -- 16	abef

[1]The Tables 49 A and 50 were inserted October 6, 1856.—A. C.

					bde^2 + 64	ace^2 + 16	
a^2cd + 36	aby + 4	abcf - 12	abcf + 6	ade^2 - 4		ad^2e + 16	acdf + 20
		abde - 6		acy - 24	acy + 24	b^2df - 15	
ab^2e + 16	cd^2ce - 84			bcd^2e - 26			- ace^2 80
	abd^2 - 24	ace^2 + 16			acrfe - 208	b^2y	
abcd - 152	ac^2d + 64		ac^2e - 96	acde + 24	ad^2 + 144		ad^2e + 60
	b^2e + 60						
moc^3 + 96	b^2cd - 40	ac(d^2) ...	ac(d^2)+ 96	ad^2	b^2y		b^2cy - 80
+ 80							
	be^4	b^4y + 16	by ...	b^2y + 24	bcy - 40		b^2y + 240
- bc^4 60		b^2ce - 10	b^2ce + 90	b^2ce - 20	+ 60 bcr^2	bcdy	bcdy + 60
						bcde	
		b^2cd ...	- b^2cd 80	bc^2e ...	bcdy - 40	bcd	ictfe- 860
		bc^2d ...	6c^2d ...	bed^2 ...	c^2d		bd^2 + 960
		c^3	c^3	c^2d		c^2e	c^2e + 960
							- c^2d^2 320

[Continued on book p. 311 with columns 1346, 2346, 1356, 2356, 1456, 2456, 3456 in the same format.]

No. 28.

[Eight-column table listing the same monomials in a, b, c, d, e, f and giving the integer coefficients which appear in the expressions for $N, M, L, L', P, F, M', N'$.]

N.	M.	L.	L'.	P.	F.	M'.	N'.
a^2cf + 3	a^2cf + 1	a^2y^2 + 1	a^2y^2 + 3	a^2y^2 ...	a^2y - 1	ab^2f + 1	a^3y - 3
- b^2cd 2	abdf + 2	abef - 34	abef - 22	abef + 1	abef + 9	acef + 2	odef - 9
		acdf + 76		acdf - 3	acdf - 1		
abcf - 9	abe^2 - 9	ace^2 - 32	acdf - 12	ace^2 + 2		ady - 9	oe^4 + 6
					ace^2 - 18		
abde^2 + 1	adf - 9	ad^2e - 12	ace^2 + 64	ad^2e	ad^2e + 12	ad^2e + 6	jy - 2
	acde + 32						
eude + 18	ad^2 - 18	b^4y - 32	acd^2e - 36	b^2c + 2		b^2cf - 9	bcy + 1
aecd^2 - 12		b^2c + 225			b^2cf - 18		
			bcde + 64	b^2cf - 9	b^2c	bcdf - 32	bcdf + 18
		bc^2f - 12					- bd^4 15
b^2y + 6	b^2cy + 6	bcdy - 820	bc^2y - 45	bef	bcy + 12	bce^2 ...	
b^2ce - 15		bd^4 + 480	bcy - 36		bccdy + 45		edf - 12
	b^2cde ...	c^2 + 480		bcde + 31	bc^2 - 30	bd^2e - 15	
b^2cd^2 + 10		c^2d^2 - 320	bcdy + 20	bdy - 18			
	bc^2y - 15			c^2e - 18	bce - 30	cy - 18	c^2e^2 + 10
					c^2d^2 + 12		
bc^2 ...	bcd^2 + 10		bcd^2		c^2d^2 + 20	c^2de + 10	cd^2e ...
c^3	c^3d ...		c^3e			cd^2	d^4
			c^2d^2 ...				

If the coefficients of the table 14 are represented by $\frac{1}{2}A, B, \frac{1}{2}C$, viz. writing

$$A = 2(ae - 4bd + 3c^2), \qquad B = af - 3be + 2cd, \qquad C = 2(bf - 4ce + 3d^2),$$

then we have the following relations between 1234, &c. and A, B, C, viz.

```
1234 =                    Cx                 + Bx                  + A x
1235 =                                                  (table continues, see scan)
1236 =  + 6 a^2                 - 12 ab      + 16 oc - 10 b^2
1245 =  + 6 ab                  - 2 oc - 10 b^2     + 6 ac;
1246 =                          + 6 ad - 18 bc
1345 =  - 2 ac + 8 b^2          - 6 ad - 30 bc       - 2 df + 8 b^2
1256 =                          + 4 ae -- 4 bd -- 24 c^2  + 8 ae +10 bd
2345 =  + 18 ac                 - 8 ac - 40 bc;
1346 =  + 12 bc                                      + 4 be + 8 cd
2346 =  + 24 ac                 + I af + 5 be -- 18 cd   + 4 af + 20 be
1356 =                          - 80 be + 20 cd
2356 =  - I ae + 4bd + 3 c^2                         -- I bf + 4 ce + 3d^2
1456 =                          - 3 bcd             + 20 bf + 40 ce - 30 cf
2456 =  + 20 ad + 40 bc - 30 e^2  - 8 bf - 4 ce
3456 =                                              + 4 bf + 8 ce + 6 d^2
         + 4 ae + 8bd+ 6 c^2      + 4 bf - 4 ce - 24 c^2  + 24 cf
                                  - 6 cf - 30 cfe          + 12 cfe
         + 4 af + 20 be
         + 4 be + 8 cd           + 8 cf - 18 de            + 18 df
         + 8 bf + 10 ce          - 2 df - 10 e^2            - 2 df + 8 e^2
         + 6 ce                  -12 ef                    + 6 ef
         + 6 cf                                             + 6 f^2

         + 16 df - 10 e^2
```

and the following relations between L, L', &c. and A, B, C, viz.

$N =$	Cx	$+Bx$	$+Ax$
$M =$	$-3ac+3b^2$	$+3ad-3bc$	$-1ae+1be$
	$-3ad+3bc$		$-1af+1cd$
$L =$	$+11ae+28bd-39c^2$	$+3ae-3d^2$	$+11bf+28ce-39d^2$
	$-7af+4bd+3c^2$		$-7bf+4ce+3d^2$
$L' =$		$+1af-75be+74cd$	$+1bf+2ce-3d^2$
$2P =$	$-7ae-2bd+8c^2$	$+3af+15be-18cd$	$+3bf-6ce+3d^2$
	$+3ae-6bd+3c^2$	$+3be-3cd$	
$F =$		$-1af+1cd$	$y-3+3de$
	$-1af+1cd$	$+3bf-3d^2$	
$M' =$		$+3cf-3de$	$-3df+3e^2$
$N' =$	$-1bf+1ce$		

We have also the following relations between L, L', &c. and a, b, c, d, e, f, viz.

$$
\begin{aligned}
aP - bM + cN &= 0,\\
aM' + bP' - 2cM + 3dN &= 0,\\
aN' + 2bM' - cL' + 3eN &= 0,\\
3bN' - dL' + 2eM + fN &= 0,\\
3cN' - 2dM' + eP' + fM &= 0,\\
dN' - eM' + fP &= 0.
\end{aligned}
$$

The quartinvariant No. 19 [G] is equal to

$$-AC + B^2,$$

i.e. it is in fact equal to 4 into the discriminant of the quintic No. 14, [A].

The octinvariant No. 25 [Q] is expressible in terms of the coefficients of Nos. 14 and 16, viz. A, B, C, as before, and $\frac{1}{2}a, \beta, \gamma, \frac{1}{2}\delta$ the coefficients of No. 16, [D], i.e.

$$\begin{aligned}
\alpha &= 3(ace - ad^2 - b^2e + 2bcd - c^3),\\
\beta &= acf - ade - b^2f + bd^2 + bce - c^2d,\\
\gamma &= adf - ae^2 - bcf + bde + c^2e - cd^2,\\
\delta &= 3(bdf - be^2 + 2cde - c^2f - d^3),
\end{aligned}$$

then No. 25 is equal to

$$\begin{vmatrix} A & B & C \\ \alpha & \beta & \gamma \\ \beta & \gamma & \delta \end{vmatrix}.$$

The value of the discriminant No. 26, [Q'], is

$$(\text{No. } 19)^2 - 128\,\text{No. } 25, \qquad [\text{that is } Q' = G^2 - 128Q.]$$

We have also an expression for the discriminant in terms of L, L', &c., viz. three times the discriminant No. 26 is equal to

$$[\text{or say } 3Q' =]\ \ LL' + 64MM' - 64NN',$$

a remarkable formula, the discovery of which is due to Mr. Salmon.

It may be noticed, that in the particular case in which the quintic has two square factors, if we write

$$(a, b, c, d, e, f \mid x, y)^5 \ = \ \{(p, q, r \mid x, y)^2\}^2 \cdot (\lambda, \mu \mid x, y),$$

then

$$\begin{aligned}
a &= \lambda p^2, & b &= \tfrac{1}{4}(p^2\mu + 4pq\lambda), & c &= \tfrac{1}{2}(q^2 + pr)\lambda + 2pq\mu,\\
f &= 5r^2\mu, & e &= r^2\lambda + 4qr\mu, & d &= 2qr\lambda + (2q^2 + pr)\mu;
\end{aligned}$$

and these values give

$$\begin{aligned}
P &= K(6q^2 - pr), & P' &= K(10q^2 - 15pr),\\
M &= K \cdot 10pq, & M' &= K \cdot 10qr,\\
N &= K \cdot 5p^2, & N' &= K \cdot 6r^3,
\end{aligned}$$

where the value of K is

$$\tfrac{1}{8}(p\mu^2 - 2q\mu\lambda + r\lambda^2)\,(pr - q^2)^3.$$

The table No. 29 is the invariant of the twelfth degree of the quintic, given in its simplest form, i.e. in a form not containing any power higher than the fourth of the leading coefficient a: this invariant was first calculated by M. Faà de Bruno.

No. 29. [See U. No. 29, p. 294.]

[Reference back to the corresponding entry in the U-table of paper 143.]

The tables Nos. 30 to 35 relate to a sextic. No. 30 is the sextic itself; No. 31 the quadrinvariant; Nos. 32 and 33 the quadricovariants (the latter of them the Hessian); No. 34 is the quartinvariant or catalecticant; and No. 35 is the sextinvariant in its best form, i.e. a form not containing any power higher than the second of the leading coefficient a.

No. 30.

$$(a+1,\ b+6,\ c+15,\ d+20,\ e+15,\ f+6,\ g+1 \mid x, y)^6.$$

No. 31.

$$\left.\begin{array}{l} ag+1, \\ bd-6, \\ ce+15, \\ d^2-10, \end{array}\right. \quad \text{coefficient of } ag^0e^0; \qquad \text{sum } \pm 16.$$

No. 32.

$ae+1,$ $bd-4,$ $c^2+3,$	$af+2,\ bg+1,$ $be-6,\ cf-6,$ $cd+4,\ de+4,$	$ag+1,\ cg+1$ $ce-9,\ df-4$ $d^2+8,\ e^2+3$
±4	$\pm6,\ \pm6$	±9

$(\diamond x, y)^2$.

No. 33.

$ac+1$	$ad+4$	$ae+6$	$af+4$	$ag+1$	$bg+4$	$cg+6$	$dg+4$	$eg+1$
	$bc-4$		$be+16$	$bf+14$	$cf+16$		$df+4$	$ef-4$
b^2-1		$bd+4$	$cd-20$	$ce+5$		$df+4$	e^2-10	f^2-1
		c^2-10		d^2-20	$de-20$			
±1	±4	±10	±20	±20	±20	±10	±4	±1

$(\diamond x, y)^4$.

No. 34.

[The quartinvariant or catalecticant of the sextic: a long list of monomial-coefficient pairs which sum, as printed in the source, to ±12. The table is reproduced below in OCR-skeleton form.]

```
a^2dg^2 + 1   acd^2y - 42   bcdfg + 60   bdf^2 + 24   c^2eg + 12   c^2y - 27   - c^2d^2gr 8
acde^2 + 12                                           - 24 bd^2e^2
a^2defg -- 6                                                        bdy^2 +24    cdy^2 - 27
a^2y^2 + 4    ad^2g -20      bcfeg^2 + 4 c^2gr + 12   c^2e^2/ +66   c^2eg^2 - 24                    c^2def
a^2bf^2 + 4   a^2cef + 24    bg + 12     c^2cg^2 +24 - cf^2 - 24    cy^2 - 24                      c^2y - 6
a^2y^2 - 3    abce^2 - 8     by^2 + 8    cd^2eg^2 -3 -39 c^2d^2e^2  cd^2g - 24                      c^2cd^2
ab^2cdg^2 - 6 b^2gr - 12     b^2cdy +30  cd^2eg^2 -39  cd^2e^2 + 36
abce/g+ 18    by^2 + 8       b^2ce^2 -24                            c^2y - g
abc^2y -12

ab^2dy^2 + 12 b^2ce^2 -12
abdee^2 -18   b^2dy^2 - 24
abe^2y + 6
ac^2d^2 + 4   b^2cdy + 60
ac^2eg -24
ac^2dy -18    b^2c^2 - 27
ab/^2 + 30    bcefg + 6
acd^2eg + 54  bcdef - 42
acrdy -12
            ± 12                                                                                ± 565
```

[Layout preserved from the OCR; the column totals are ±12 and ±565 in the printed table; sub/superscript glitches reflect the OCR not the manuscript.]

The sextinvariant may be thus represented by means of a determinant of the sixth order and of the quadrinvariant and quartinvariant:

$$6 \times \text{No. } 35 \;=\; \left|\begin{array}{ccccccc} a, & 2b, & 3c, & 4d, & e, & \cdot & \cdot \\ \cdot & a, & 2b, & 3c, & 4d, & e, & \cdot \\ \cdot & \cdot & a, & 2b, & 3c, & 4d, & e \\ a, & 4b, & 3c, & 2d, & e, & \cdot & \cdot \\ \cdot & a, & 4b, & 3c, & 2d, & e, & \cdot \\ \cdot & \cdot & a, & 4b, & 3c, & 2d, & e \end{array}\right| \;+\; \tfrac{1}{4}(ag-6bf+15ce-10d^2)\left|\begin{array}{cccc} a, & b, & c, & d \\ b, & c, & d, & e \\ c, & d, & e, & f \\ d, & e, & f, & g \end{array}\right|.$$

The tables Nos. 36 and 37 relate to a septimic. No. 36 is the septimic itself; No. 37 the quartinvariant.

No. 36.

$$(a+1,\ b+7,\ c+21,\ d+35,\ e+35,\ f+21,\ g+7,\ h+1 \mid x,y)^7.$$

No. 37.

[The quartinvariant of the septimic: monomial expansion totalling ± 2223 *as in the source. Sample terms:]*

$$\begin{array}{lllll} a^2h^2-1, & bd^2h-40, & bdeg-50, & bdf^2-360, & \\ abgh+14, & b^2fh+240, & acgh-18, & c^2eg-360, & \\ acfh-24, & cf^2-81, & afg^2+10, & cd^2g+240, & \\ adeh+60, & cdef+990, & adfg-40, & cef-600, & \\ adf-24, & cf^2-600, & ae^2g-25, & d^2e^2+375, & \\ b^2fh+234, & by+60, & bcfg, & bceh, & \text{etc.} \end{array}$$

Total: ± 2223.

The tables Nos. 38 to 45 relate to the octavic. No. 38 is the octavic itself; No. 39 the quadrinvariant; Nos. 40, 41 and 42 are the quadricovariants, the last of them being the Hessian; No. 43 is the cubinvariant; No. 44 the quartinvariant, and No. 45 the quintinvariant, which is also the catalecticant.

No. 38.

$$(a+1,\ b+8,\ c+28,\ d+56,\ e+70,\ f+56,\ g+28,\ h+8,\ i+1 \mid x,y)^8.$$

No. 39.

$$ai+1,\quad bh-8,\quad cy+28,\quad df-56,\quad e^2+35;\quad \text{sum}\ \pm 64.$$

Nos. 40, 41, 42.

[Tabulations of the quadricovariants $(\diamond x,y)^2$, $(\diamond x,y)^4$, $(\diamond x,y)^6$ *of the octavic; the column-sums are* ±16, ±20, ±35, ±20, ±16 *for No. 40;* ±4, ±12, ±30, ±48, ±58, ±48, ±30, ±12, ±4 *for No. 41;* ±1, ±6, ±21, ±70, ±105, ±210, ±189, ±210, ±105, ±70, ±21, ±6, ±1 *for No. 42.]*

No. 43. (The cubinvariant.)

```
aei + 1     acgi - 1     bcgh + 3   cdei -2
afh -- 4    ach^2 + 1    bdei + 1   cdeh - 23
ag^2 + 3                            cdfh + 27
bdi -- 4    adfi + 3     bdfh - 10
                                    cdfg + 19
beh + 12    adgh - 3     bdg^2 + 9
                                    ce^2i - 21
b^2 - 8     -aei - 1 2   be^2 i + 11
                                    cefh + 12
c^2i + 3    aefh + 1     befg - 23
                                    cy^2 -- 21
cdh - 8     aeg^2 + 3    bfh + 12
                                    d^2eg - 13
ceg - 22                 d^2gi + 3
                                    d^2y - 32
cy^2 + 24   af^2g - 2    e^2fh + 9  - 10
d^2g + 24                de^2f + 10
def -- 36   b^2gi + 1    e^3f - 12
e^2 + 15    b^2h^2 - 1
                                    ± 82

            bcfi - 3
```

No. 44 (the quartinvariant) and No. 45 (the quintinvariant / catalecticant).

[Both invariants are given by long tables of monomial-coefficient pairs; the column totals are ±147 for No. 44 and ±56 for No. 45 (printed at the foot of the tables in the source; some sub-totals are damaged in the scan).]

If we write

$$\text{No. } 39 = I, \quad \text{No. } 43 = J, \quad \text{No. } 44 = K, \quad \text{No. } 45 = L,$$

then the determinant called the lambdaic, viz.

$$\begin{vmatrix} a & b & c & d & e-12\lambda \\ b & c & d & e+3\lambda & f \\ c & d & e-2\lambda & f & g \\ d & e+3\lambda & f & g & h \\ e-12\lambda & f & g & h & i \end{vmatrix}$$

is equal to

$$L + 4\lambda K + 24\lambda^2 J + 18\lambda^3 I - 2592\lambda^5.$$

Nos. 46 to 48 relate to the nonic. No. 46 is the nonic itself; Nos. 47 and 48 are the two quartinvariants, each of them in its best form, viz. No. 48 does not contain a^2, and No. 47 does not contain aci^3, the leading term of No. 48. The nonic is the lowest quantic with two quartinvariants.

No. 46.

$$(a+1,\ b+9,\ c+36,\ d+84,\ e+126,\ f+126,\ g+84,\ h+36,\ i+9,\ j+1 \mid x, y)^9.$$

Nos. 47, 48.

[The two quartinvariants of the nonic, listed as monomial-coefficient tables. The column-sums for No. 47 are ±41650; for No. 48 they are ±698. Sample leading terms:]

```
No. 47.                                 No. 48.

a^2y^3 - 1      bfh - 720              ay^3 ...          bfgh + 70
abij + 18       b/g^2 + 432            abij              bfg^2 - 45
aei^2           c^2fj + 1728           acy^3 + 2         c^2ij + 27
                                       achj - 2          c^2gi - 52
achj - 72       c^2gi - 720
                                       adgj + 7          c^2y^2 + 25
adgj + 168      c^2y^2 2160            adhi - 7          cdej - 45
adhi            cdej + 4608            aej^2 - 5         cdfi + 23
aei^2 - 108     cdfi - 576             aei^2 - 22        crfgr/h + 22
- ciegi 576     cdfg^2 + ...           adi^2 + 27        c^2i + 70
aeh^2 + 432     ce^2y ...              af^2i + 25        c^2/h - 127
afy + 540       cefh - 2592            b^2fgh - 45       cef + 32
                                             + 20        c/^2 + 25
b^2afgh -720    ce^2g - 5760
        +320    cy^3 + 4320            b^2fhj + 2        cf^2y + 20
                d^2j + 320             bcg^2 - 7         d^2ei - 45
b^2b^2ij                                          +7    d^2y/h + 32
       -- 81    d^2ei - 720            bcfhj
                                                         d^2g^2 + 47
bcgj + 648      dy^2 - 5760            bdfj -22          d^2y^2 + 85
bchi + 648      d^2g^2 - 1536          bdgi + 74
                                                         de^2 h + 25
bdfj - 676      defg + 14688           bdh^2 - 52
bdgi + 792      deyhh + 4320           begh + 23         dy^2 - 50
bdh^2 - 1728                           be^2 j + 25       - e^2 f/  50
```

```
beg^2 + 540    cd^2f^2 - 8640
                                        bfy - 73        ey^2 + 30

                                          ± 41650                                        ± 698
```

Nos. 49, [49 A] and 50 relate to the dodecadic. No. 49 is the dodecadic itself; [No. 49 A, inserted in this place, but originally printed in the Fifth Memoir on Quantics, is the dodecadic quadricovariant]. No. 50 is the cubinvariant. [The numerical coefficients in this last table as originally printed in the Third Memoir were altogether erroneous, and the table as here printed is in fact the table No. 50 bis of the Fifth Memoir on Quantics.]

No. 49.

$(a{+}1,\ b{+}12,\ c{+}66,\ d{+}220,\ e{+}495,\ f{+}792,\ g{+}924,\ h{+}792,\ i{+}495,\ j{+}220,\ k{+}66,\ l{+}12,\ m{+}1 \mid x, y)^{12}$.

No. 49 A.

[The dodecadic quadricovariant, given as a thirteen-column tabulation $(\diamond x, y)^{10}$ *with column-sums* ±16, ±60, ±189, ±450, ±810, ±1140, ±1270, ±1140, ±810, ±450, ±189, ±60, ±16.*]*

No. 50 (Cubinvariant of the dodecadic, total ±2200).

```
agm + 1      cfj^2 - 54     dhi + 270
ahl - 6      cgk + 24       eg^2 -135
aik + 15                    efj +270
aj^2 - 10    c/fy + 150     eg^2 i + 495
bfm - 6      cf^2 - 135     eie - 540
                             f^2 i - 540
bgl + 30     d^2 m - 10
bhb -54      d^2j + 30      fgh + 720
bij + 30     dfk + 150       (g^3 -280
             cfgy -430
cem +15

                                              ± 2200
```

Resuming now the general subject:

54. The simplest covariant of a system of quantics of the form

$$(\diamond x, y, \ldots)^m$$

(where the number of quantics is equal to the number of the facients of each quantic) is the functional determinant or Jacobian, viz. the determinant formed with the differential coefficients or derived functions of the quantics with respect to the several facients.

55. In the particular case in which the quantics are the differential coefficients or derived functions of a single quantic, we have a corresponding covariant of the single quantic, which covariant is termed the *Hessian*; in other words, the Hessian is the determinant formed with the second differential coefficients or derived functions of the quantic with respect to the several facients.

56. The expression, an *adjoint linear form*, is used to denote a linear function $\xi x + \eta y + \ldots$, or in the notation of quantics $(\xi, \eta, \ldots \mid x, y, \ldots)$, having the same facients as the quantic or quantics to which it belongs, and with indeterminate coefficients $(\xi, \eta, \ldots)$. The invariants of a quantic or quantics, and of an adjoint linear form, may be considered as quantics having $(\xi, \eta, \ldots)$ for facients, and of which the coefficients are of course functions of the coefficients of the given quantic or quantics. An invariant of the class in question is termed a *contravariant* of the quantic or quantics. The idea of a contravariant is due to Mr. Sylvester.

In the theory of binary quantics, it is hardly necessary to consider the contravariants; for any contravariant is at once turned into an invariant by writing $(y, -x)$ for (ξ, η).

57. If we imagine, as before, a system of quantics of the form

$$(\diamond x, y, \ldots)^m$$

where the number of quantics is equal to the number of the facients in each quantic, the function of the coefficients, which, equated to zero, expresses the result of the elimination of the facients from the equations obtained by putting each of the quantics equal to zero, is said to be the *Resultant* of the system of quantics. The resultant is an invariant of the system of quantics.

And in the particular case in which the quantics are the differential coefficients, or derived functions of a single quantic with respect to the several facients, the resultant in question is termed the *Discriminant* of the single quantic; the discriminant is of course an invariant of the single quantic.

58. Imagine two quantics, and form the equations which express that the differential coefficients, or derived functions of the one quantic with respect to the several facients, are proportional to those of the other quantic. Join to these the equations obtained by equating each of the quantics to zero; we have a system of equations, one of which is contained in the others, and from which therefore the facients may be eliminated. The function which, equated to zero, expresses the result of the elimination is an invariant which (from its geometrical signification) might be termed the *Tactinvariant* of the two quantics, but I do not at present propose to consider this invariant except in the particular case where the system consists of a given quantic and of an adjoint linear form. In this case the tactinvariant is a contravariant of the given quantic, viz. the contravariant termed the *Reciprocant*.

59. Consider now a quantic

$$(\diamond x, y, \ldots)^m,$$

and let the facients $x, y, \ldots$ be replaced by $\lambda x + \mu X,\ \lambda y + \mu Y,\ \ldots$; the resulting function may, it is clear, be considered as a quantic with the facients (λ, μ) and of the form

$$(\diamond x, y, \ldots)^m_{\text{eman.}} \; .$$

The coefficients of this quantic are termed *Emanants*, viz., excluding the first coefficient, which is the quantic itself (but which might be termed the 0-th emanant), the other coefficients are the first, second, and last or ultimate emanants. The ultimate emanant is, it is clear, nothing else than the quantic itself, with $(X, Y, \ldots)$ instead of $(x, y, \ldots)$ for facients; the penultimate emanant is, in like manner, obtained from the first emanant by interchanging $(x, y, \ldots)$ with $(X, Y, \ldots)$, and similarly for the other emanants. The facients $(X, Y, \ldots)$ may be termed the facients of emanation, or simply the new facients. The theory of emanation might be presented in a more general form by employing two or more sets of emanating facients; we might, for example, write $\lambda x + \mu X + \nu X',\ \lambda y + \mu Y + \nu Y',\ \ldots$ for $x, y, \ldots$, but it is not necessary to dwell upon this at present.

The invariants, in respect to the new facients, of any emanant or emanants of a quantic (i.e. the invariants of the emanant or emanants, considered as a function or functions of the new facients), are, it is easy to see, covariants of the original quantic, and it is in many cases convenient to define a covariant in this manner; thus the Hessian is the discriminant of the second or quadric emanant of the quantic.

60. If we consider a quantic

$$(a, b, \ldots \mid x, y, \ldots)^m,$$

and an adjoint linear form, the operative quantic

$$(\partial_a, \partial_b, \ldots \mid \xi, \eta, \ldots)^k$$

(which is, so to speak, a contravariant operator) is termed the *Evector*. The properties of the evector have been considered in the introductory memoir, and it has been in effect shown that the evector operating upon an invariant, or more generally upon a contravariant, gives rise to a contravariant. Any such contravariant, or rather such contravariant considered as so generated, is termed an *Evectant*. In the case of a binary quantic,

$$(a, b, \ldots \mid x, y)^m,$$

the covariant operator

$$(\partial_a, \partial_b, \dots \mid y, -x)^k$$

may, if not with perfect accuracy, yet without risk of ambiguity, be termed the Evector, and a covariant obtained by operating with it upon an invariant or covariant, or rather such covariant considered as so generated, may in like manner be termed an Evectant.

61. Imagine two or more quantics of the same order,

$$(a, b, \dots \mid x, y)^m, \qquad (\alpha, \beta, \dots \mid x, y)^m, \quad \dots$$

we may have covariants such that for the coefficients of each pair of quantics the covariant is reduced to zero by the operators

$$a\partial_\alpha + b\partial_\beta + \dots,$$
$$\alpha\partial_a + \beta\partial_b + \dots.$$

Such covariants are called *Combinants*, and they possess the property of being invariantive, quoad the system, i.e. the covariant remains unaltered to a factor près, when each quantic is replaced by a linear function of all the quantics. This extremely important theory is due to Mr. Sylvester.

Proceeding now to the theory of ternary quadrics and cubics.

First for a ternary quadric, we have the following tables:

Covariant and other Tables, Nos. 51 to 56 (a ternary quadric).

No. 51.

The quadric is represented by

$$(a, b, c, f, g, h \mid x, y, z)^2,$$

which means

$$ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy.$$

No. 52.

The first derived functions (omitting the factor 2) are

$$(a, h, g \mid x, y, z),$$
$$(h, b, f \mid x, y, z),$$
$$(g, f, c \mid x, y, z).$$

No. 53.

The operators which reduce a covariant to zero are

$$(\,h,\ b,\ 2f \mid \partial_g, \partial_h, \partial_c\,) - z\partial_y,$$
$$(\,a,\ 2h,\ g \mid \partial_h, \partial_b, \partial_f\,) - y\partial_z,$$
$$(\,g,\ 2f,\ c \mid \partial_a, \partial_h, \partial_g\,) - z\partial_x,$$
$$(\,a,\ h,\ 2g \mid \partial_h, \partial_b, \partial_f\,) - x\partial_y,$$
$$(\,2h,\ b,\ f \mid \partial_a, \partial_h, \partial_g\,) - x\partial_z,$$
$$(\,2g,\ f,\ c \mid \partial_a, \partial_h, \partial_g\,) - z\partial_x.$$

No. 54.

The evector is

$$(\partial_a, \partial_b, \partial_c, \partial_f, \partial_g, \partial_h \mid \xi, \eta, \zeta)^2.$$

No. 55.

The discriminant is

$$\begin{vmatrix} a, & h, & g \\ h, & b, & f \\ g, & f, & c \end{vmatrix},$$

which is equal to

$$abc - af^2 - bg^2 - ch^2 + 2fgh.$$

No. 56.

The reciprocant is

$$\begin{vmatrix} 0, & \xi, & \eta, & \zeta \\ \xi, & a, & h, & g \\ \eta, & h, & b, & f \\ \zeta, & g, & f, & c \end{vmatrix},$$

which is equal to

$$(bc - f^2,\ ca - g^2,\ ab - h^2,\ gh - af,\ hf - bg,\ fg - ch \mid \xi, \eta, \zeta)^2.$$

The discriminant is, it will be noticed, the same function as the Hessian. The reciprocant is the evectant of the discriminant. The covariants are the quadric itself and the discriminant; the reciprocant is the only contravariant.

Next, for a ternary cubic, we have the following Tables.

Covariant and other Tables, Nos. 57 to 70 (a ternary cubic).

No. 57.

The cubic is $U =$

$$(a, b, c, f, g, h, i, j, k, l \mid x, y, z)^3,$$

which means

$$ax^3 + by^3 + cz^3 + 3fy^2z + 3gz^2x + 3hx^2y + 3iyz^2 + 3jzx^2 + 3kxy^2 + 6lxyz.$$

No. 58.

The first derived functions (omitting the factor 3) are

$$(a, k, g, l, j, h \mid x, y, z)^2,$$
$$(h, b, i, f, l, k \mid x, y, z)^2,$$
$$(j, f, c, i, g, l \mid x, y, z)^2.$$

The second derived functions (omitting the factor 6) are

$$(a, h, j \mid x, y, z),$$
$$(k, b, f \mid x, y, z),$$
$$(g, i, c \mid x, y, z),$$
$$(l, f, i \mid x, y, z),$$
$$(j, l, g \mid x, y, z),$$
$$(h, k, l \mid x, y, z).$$

No. 59.

The operators which reduce a covariant to zero are

$$\begin{gathered}(j,3f,c,2i,g,2l \mid \partial_h,\partial_b,\partial_i,\partial_f,\partial_l,\partial_k) - y\partial_z,\\ (a,k,3g,2l,2j,h \mid \partial_j,\partial_f,\partial_c,\partial_i,\partial_g,\partial_l) - z\partial_z,\\ (3h,b,i,f,2l,2k \mid \partial_a,\partial_k,\partial_g,\partial_l,\partial_j,\partial_h) - x\partial_y,\\ (h,b,3i,2f,2l,k \mid \partial_j,\partial_f,\partial_c,\partial_i,\partial_g,\partial_l) - z\partial_y,\\ (3j,f,c,i,2g,2l \mid \partial_a,\partial_k,\partial_g,\partial_l,\partial_j,\partial_h) - x\partial_z,\\ (a,k,g,2l,j,2h \mid \partial_k,\partial_b,\partial_i,\partial_f,\partial_l,\partial_k) - y\partial_x.\end{gathered}$$

No. 60.

The evector is

$$(\partial_a,\partial_b,\partial_c,\partial_f,\partial_g,\partial_h,\partial_i,\partial_j,\partial_k,\partial_l \mid \xi,\eta,\zeta)^3.$$

No. 61.

The Hessian is $HU =$

$$\begin{vmatrix} (a,h,j \mid x,y,z), & (h,k,l \mid x,y,z), & (j,l,g \mid x,y,z) \\ (h,k,l \mid x,y,z), & (k,b,f \mid x,y,z), & (l,f,i \mid x,y,z) \\ (j,l,g \mid x,y,z), & (l,f,i \mid x,y,z), & (g,i,c \mid x,y,z) \end{vmatrix},$$

which is equal to

$$(\,agh-1,\;\; bhi-1,\;\; -cij\,1,\;\; bch-1,\;\; acf-1,\;\; abg-1,\;\; bcj-1,\;\; ach-1,\;\; abi-1,\;\; ahc-1\,;\;\ldots \mid x,y,z)^6.$$

[The table runs across ten columns with $\pm3,\pm3,\pm3,\pm6,\pm5,\pm5,\pm5,\pm5,\pm5,\pm9$ as column sums. The body of the table is a long enumeration of monomial-coefficient pairs which, by the size and density of the printing, cannot be faithfully retyped without the scan; see the original.]

No. 62.

The quartinvariant is $S =$

$$\begin{array}{ll} abcl-1, & fjy+1, \\ abgi+1, & f^2h+3, \\ acfk+1, & fgjk-1, \\ af^2j-1, & fhiy-1, \\ afil+1, & fjl^2-2, \\ afi^2-1, & ghi^2+1, \\ aghk-1, & ghki-1, \\ bchj+1, & ghl^2-2, \\ bfh-1, & h^2i^2+1, \\ bgji+1, & hil^2-2, \\ bgj^2-1, & i^2jl+3, \\ cf^2h-1, & l^4+1, \\ cf^2k^2+1, & \\ cjk^2-1, & \end{array}$$

Total ±16.

No. 63.

The sextinvariant is $T =$ *[a very long table giving, in source-printing, ± 871 as the total of all monomial-coefficient pairs. The discovery of the invariants S and T is due to Aronhold; the developed expressions were first obtained by Mr. Salmon.]*

No. 64.

There is an octicovariant for which we may take

$$\Theta U = \begin{vmatrix} & \partial_x HU & \partial_y HU & \partial_z HU \\ \partial_x HU & \frac{1}{6}\partial_x^2 U & \frac{1}{6}\partial_x\partial_y U & \frac{1}{6}\partial_x\partial_z U \\ \partial_y HU & \frac{1}{6}\partial_y\partial_x U & \frac{1}{6}\partial_y^2 U & \frac{1}{6}\partial_y\partial_z U \\ \partial_z HU & \frac{1}{6}\partial_z\partial_x U & \frac{1}{6}\partial_y\partial_z U & \frac{1}{6}\partial_z^2 U \end{vmatrix},$$

or else

$$\Theta_{,}U = \begin{vmatrix} & \frac{1}{2}\partial_x U & \frac{1}{2}\partial_y U & \frac{1}{2}\partial_z U \\ \frac{1}{3}\partial_x U & \partial_x^2 HU & \partial_x\partial_y HU & \partial_x\partial_z HU \\ \frac{1}{3}\partial_y U & \partial_y\partial_x HU & \partial_y^2 HU & \partial_y\partial_z HU \\ \frac{1}{3}\partial_z U & \partial_z\partial_x HU & \partial_z\partial_y HU & \partial_z^2 HU \end{vmatrix},$$

or else, what I believe is more simple, a function $\Theta_{,,}U$, which is a linear function of the last-mentioned two functions.

The relations between $\Theta U,\ \Theta_{,}U,\ \Theta_{,,}U$ are

$$-\Theta_{,}U + 4\Theta U = T \cdot U^2 - 24S \cdot U \cdot HU,$$
$$\Theta_{,,}U + 2\Theta U = T \cdot U^2 - 10S \cdot U \cdot HU.$$

I have not worked out the developed expressions.

No. 65.

The cubicontravariant is $PU =$ *[an extensive table of monomial-coefficient pairs in $a, b, c, f, g, h, i, j, k, l$ multiplied by appropriate products of ξ, η, ζ. The column-sums in the printed table are $\pm 3, \pm 3, \pm 7, \pm 7, \pm 7, \pm 7, \pm 7, \pm 7, \pm 13$.]*

```
bcl - 1     acl -- 1     abl - 1     ack + 1    abi + 1    bcj + 1    abg + 1    bch +1     acf + 1    abc -- 1
a^2 - 1     b^2 - 1      bgi + 1     agi + 1    afk + 1    af^2y - 2 - 1 af^2 + 1  bgl + 1  ai^2 - 1  ahi + 1
b^2         cfk + 1      chj + 1     bhj + 1    acj + 1    bgh - 2    c/h - 2    aik - 2
                                                                              - 2 c^2h +1  bgj + 1

fg - 1      g^2 h - 1    fh^2 - 1    chk - 1    bil + 1    ckl + 1    bf - 1     chk - 1  cjk - 2    chk + 1
fil + 1     gxl + 1      ckl + 1     fy + 2     fhl + 3    ffk + 3    fhj - 1    fk^2 + 2 fgj - f    fgh + 3
ihx - 1     ij - 1       jh^2 - 1    fky + 3    fk - f     fy - 1     ghk - 1    ki - 1   fk + 2    fjl -4

            jhi gjk - 1  gki + 1   gki - 1      hhi + 2    //h - 1    ghi - 1    ghj - 4
            jy - 1       hki - 1   hi^2 + 2     hl^2 - 2   fl^2 - 2   gl^2 - 2

            jl^2 - 2     kl^2 - 2  il^2 - 2     jkl + 3    jhl + 3    jjl + 3    jjk + 3

                                                                                 l^4 + 4
```

[Layout as in the OCR; the table is followed by a parenthetical $(\xi, \eta, \zeta)^3$ marker.]

No. 66.

The quintic contravariant is $QU =$ *[an even longer table than No. 65; the column-sums in the printed table are ± 145, ± 145, ± 145, ± 282, ± 282, ± 282, ± 282, ± 282, ± 282, ± 486. The form is again a polynomial in (ξ, η, ζ) of degree 5.]*

No. 67.

The reciprocant is $FU = (* \mid \xi, \eta, \zeta)^6$, given as a long monomial table in $a, b, c, f, g, h, i, j, k, l$ multiplied by appropriate powers of ξ, η, ζ. The column-totals: $\pm 9, \pm 9, \pm 9, \pm 54, \pm 54, \pm 54, \pm 54, \pm 54, \pm 54, \pm 135, \pm 135, \pm 135, \pm 135,$

The preceding Tables contain the complete system [not so] of the covariants and contravariants of the ternary cubic, i.e. the covariants are the cubic itself U, the quartinvariant S, the sextinvariant T, the Hessian HU, and an octicovariant, say ΘU; the contravariants are the cubicontravariant PU, the quinticontravariant QU, and the reciprocant FU.

The contravariants are all of them evectants, viz. PU is the evectant of S, QU is the evectant of T, and the reciprocant FU is the evectant of QU, or what is the same thing, the second evectant of T.

The discriminant is a rational and integral function of the two invariants; representing it by R, we have $R = 64\,S^3 - T^2$.

If we combine U and HU by arbitrary multipliers, say α and 6β, so as to form the sum $\alpha U + 6\beta HU$, this is a cubic, and the question arises, to find the covariants and contravariants of this cubic; the results are given in the following Table.

No. 68.

$$
\begin{aligned}
\alpha U + 6\beta HU &= \alpha U + 6\beta HU. \\
H(\alpha U + 6\beta HU) &= (0,\ 2S,\ T,\ 8S^2 \mid \alpha, \beta)^3 \cdot U \\
&\quad + (1,\ 0,\ -12S,\ -2T \mid \alpha, \beta)^3 \cdot HU. \\
P(\alpha U + 6\beta HU) &= (1,\ 0,\ 12S,\ 4T \cdot 8 \mid \alpha, \beta)^3 \cdot PU \\
&\quad + (0,\ 1,\ 0,\ -4S^2 \mid \alpha, \beta)^3 \cdot QU. \\
Q(\alpha U + 6\beta HU) &= (0,\ 60S,\ 30T,\ 0,\ -120TS,\ -240S^2 \mid \alpha, \beta)^5 \cdot PU \\
&\quad + (1,\ 0,\ 0,\ 10T,\ 240S^2,\ 24TS^2 \mid \alpha, \beta)^5 \cdot QU. \\
S(\alpha U + 6\beta HU) &= (S,\ T,\ 24S^2,\ 4TS,\ T^2 - 4S^3 \mid \alpha, \beta)^4. \\
T(\alpha U + 6\beta HU) &= (T,\ 96S^2,\ 60TS,\ 20T^2,\ 240TS^2 + 48T^2S + 4608S^4,\ -8T^3 + 576TS^3 \mid \alpha, \beta)^6. \\
R(\alpha U + 6\beta HU) &= [(1,\ 0,\ -24S,\ -8T,\ -48S^2 \mid \alpha, \beta)^4]^3 \cdot PU. \\
F(\alpha U + 6\beta HU) &= FU \cdot (1,\ 0,\ -24S,\ -8T,\ -48S^2 \mid \alpha, \beta)^4 \\
&\quad + (PU)^2 \cdot (0,\ 24,\ 0,\ 0,\ -48T \mid \alpha, \beta)^4 \\
&\quad + PU \cdot QU \cdot (0,\ 0,\ 24,\ 0,\ 96S \mid \alpha, \beta)^4 \\
&\quad + (QU)^2 \cdot (0,\ 0,\ 0,\ 8,\ 0 \mid \alpha, \beta)^4.
\end{aligned}
$$

We have, in like manner, for the covariants and contravariants of the cubic $6\alpha PU + \beta QU$, the following Table.

No. 69.

$$
\begin{aligned}
6\alpha PU + \beta QU &= 6\alpha PU + \beta QU. \\
H(6\alpha PU + \beta QU) &= (-2T,\ 48ST,\ 18TS,\ T^2 + 16S^3 \mid \alpha, \beta)^3 \cdot PU \\
&\quad + (8S,\ T,\ -8S^2,\ -TS \mid \alpha, \beta)^3 \cdot QU. \\
P(6\alpha PU + \beta QU) &= (32S^2,\ 12TS,\ T^2 + 32S^3,\ 4TS^2 \mid \alpha, \beta)^3 \cdot U \\
&\quad + (4T,\ 96S^2,\ 12TS,\ T^2 - 32S^3 \mid \alpha, \beta)^3 \cdot HU. \\
Q(6\alpha PU + \beta QU) &= \big(384TS^3 + 120T^2S + 7680S^4,\ 10T^2 + 320TS^2 \cdot (\alpha, \beta)^4,\ 480T^2S^2,\ \dots\big) \cdot U \\
&\quad + \big(-24T^2 + 4608S^2,\ 1920TS^2,\ 480T^2S,\ 30T^2 + 1920TS^2,\ \dots\big) \\
&\quad \text{[for full formula see scan].}
\end{aligned}
$$

$$S(6\alpha PU+\beta QU)=\big(T^3+192S^4,\ 128TS^2,\ 18T^2S+384S^3,\ T^3+64S^3,\ T^2S-64S^3\big)(\alpha,\beta)^4.$$

$$\begin{aligned}T(6\alpha PU+\beta QU)=\big(&8T^3+4608TS^3,\ 1920T^3S^2+73728S^4,\ 360T^3S+38400TS^4,\ 20T^4+8960T^3S,\\ &840T^4S+7680T^3S^4,\ 360T^3S+384T^4S^2+24576S^4,\ T^5-40T^3S^4+2560TS^4\big)(\alpha,\beta)^6.\end{aligned}$$

$$R(6\alpha PU+\beta QU)=[(48S^2,\ 8T,\ -96S^3,\ -24TS,\ -T^2-16S^3\mid\alpha,\beta)^4]^3\cdot R.$$

$$\begin{aligned}F(6\alpha PU+\beta QU)=\ &(192S,\ 32T,\ -384S^2,\ -96TS,\ -4T^2-64S^3\mid\alpha,\beta)^4\cdot\Theta U\\ &+(0,\ 512S^4,\ 192TS^4,\ 24T^2S,\ T^4\mid\alpha,\beta)^4\cdot U^4\\ &+(1344S^2,\ 352TS,\ 24T^2-1152S^3,\ -288TS^2,\ 20T^2S+64S^4\mid\alpha,\beta)^4\cdot U\cdot HU\\ &+(48T,\ 288S,\ 24T^2+1536S^3,\ 144TS^2\mid\alpha,\beta)^4\cdot(HU)^2.\end{aligned}$$

The tables for the ternary cubic become much more simple if we suppose that the cubic is expressed in Hesse's canonical form; we have then the following table.

No. 70.

(Hesse's canonical form for the ternary cubic.)

$$\begin{aligned}
U&=x^3+y^3+z^3+6l\,xyz.\\
S&=-l+l^4.\\
T&=1-20l^3-8l^6.\\
R&=-(1+8l^3)^3.\\
HU&=l^2(x^3+y^3+z^3)-(1+2l^3)\,xyz.\\
\Theta U&=(1+8l^3)^2\,(y^2z^2+z^2x^2+x^2y^2)\\
&\quad+(-9l^4)\,(x^4+y^4+z^4)\\
&\quad+(-2l-5l^4-20l^7)\,(x^3+y^3+z^3)\,xyz\\
&\quad+(-15l^2-78l^5+12l^8)\,x^2y^2z^2.\\
\Theta_x U&=4(1+8l^3)^2\,(y^2z^2+z^2x^2+x^2y^2)\\
&\quad+(-1+4l^3-4l^6)\,(x^3+y^3+z^3)^2\\
&\quad+(4l+100l^4+112l^7)\,(x^3+y^3+z^3)\,xyz\\
&\quad+(48l-552l^5+48l^8)\,x^2y^2z^2.\\
\Theta_{xx} U&=-2(1+8l^3)^2\,(y^2z^2+z^2x^2+x^2y^2)\\
&\quad+(1-10l^3)\,(x^3+y^3+z^3)^2\\
&\quad+(50l-180l^4+96l^7)\,(x^3+y^3+z^3)\,xyz\\
&\quad+(50l-624l^4-192l^7)\,x^2y^2z^2.\\
PU&=-4(\xi^3+\eta^3+\zeta^3)+(-1+4l^3)\,\xi\eta\zeta.\\
QU&=-4(1+10l^3)(\xi^3+\eta^3+\zeta^3)-6l^2(\xi+4l^3)\,\xi\eta\zeta.\\
FU&=4(1+8l^3)\,(\eta^2\zeta^2+\zeta^2\xi^2+\xi^2\eta^2)\\
&\quad+(\xi^3+\eta^3+\zeta^3)^2\\
&\quad-24l^2(\xi^3+\eta^3+\zeta^3)\,\xi\eta\zeta\\
&\quad-24l(1+2l^3)\,\xi^2\eta^2\zeta^2,
\end{aligned}$$

to which it is proper to join the following transformed expressions for $\Theta U, \Theta_x U, \Theta_{xx} U$, viz.

$$\begin{aligned}
\Theta U &= (1+8l^3)^2\,(y^2z^2+z^2x^2+x^2y^2) \\
&\quad + (2l-5l^4)\,U\cdot HU \\
&\quad + (-3l^2)\,(HU)^2. \\
\Theta_x U &= 4(1+8l^3)^2\,(y^2z^2+z^2x^2+x^2y^2) \\
&\quad + (-1+12l^3+4l^6)\,U^2 \\
&\quad + (-12l+16l^4)\,U\cdot HU \\
&\quad + (-12l^2)\,(HU)^2. \\
\Theta_{xx} U &= -2(1+8l^3)^2\,(y^2z^2+z^2x^2+x^2y^2).
\end{aligned}$$

The last preceding table affords a complete solution of the problem to reduce a ternary cubic to its canonical form.

[I add to the present Memoir, in the notation hereof $(a,h,c,f,g,h,i,j,k,l \mid x,y,z)^3$ for the ternary cubic, some formulae originally contained in the paper "On Homogeneous Functions of the third order with three variables," (1846), but which on account of the difference of notation were omitted from the reprint, 35, of that paper.]

Representing the determinant

$$\left|\begin{array}{lll}
ax+hy+jz, & hx+ky+bz, & jx+ly+gz, \\
hx+ky+bz, & kx+by+fz, & lx+fy+iz, \\
jx+ly+gz, & lx+fy+iz, & gx+iy+cz
\end{array}\right|$$

by

$$(A,B,C,F,G,H \mid x,y,z)^2,$$

the values of A,B,C,F,G,H (equations (10) of §35) are given by a long table, of which the first row is

$$\begin{array}{lcccccc}
 & gk & bi & cf & bc & ff & ki \\
A: & -l^2 & -r^2 & -v^2 & -fi & +ck & +bg \\
 & & & & & -2fl & -2il
\end{array}$$

and there are five further rows for B,C,F,G,H, each of comparable length; the full table is given on book pp. 333–334 in twelve rows.

Moreover writing

$$FU = \left|\begin{array}{cccccc}
a, & k, & g, & l, & j, & h \\
h, & b, & i, & f, & l, & k \\
j, & f, & c, & i, & g, & l
\end{array}\right| \quad , \quad \xi\eta\zeta \text{ insertions as above}$$

so that

$$HFU = Aa + Bh + Cc + 2Ff + 2Gg + 2Hh,$$

then the values of a,b,c,f,g,h (equations (13) of §35) are given by a further long table on book p. 334 in twelve rows, the first row being

$$a: \quad 0, \quad 2cf-2b^2, \quad 2bi-2f^2, \quad fi-bc, \quad 0, \quad 0,$$

and the remaining rows of comparable form. The table is reproduced in full in the scan; the entries follow the same pattern as in equations (10)–(13) of paper 35.

[The remainder of paper 144 continues beyond book page 334 = PDF page 350 and is taken up in the next chunk.]